Why can’t we figure out which is the “right” Calabi-Yau shape? Most string theorists blame this on the inadequacy of the theoretical tools currently being used to analyze string theory. As we shall discuss in some detail in Chapter 12, the mathematical framework of string theory is so complicated that physicists have been able to perform only approximate calculations through a formalism known as perturbation theory. In this approximation scheme, each possible Calabi-Yau shape appears to be on equal footing with every other; none is fundamentally singled out by the equations. And since the physical consequences of string theory depend sensitively on the precise form of the curled-up dimensions, without the ability to select one Calabi-Yau space from the many, no definitive experimentally testable conclusions can be drawn. A driving force behind present-day research is to develop theoretical methods that transcend the approximate approach in the hope that, among other benefits, we will be led to a unique Calabi-Yau shape for the extra dimensions. We will discuss progress along these lines in Chapter 13.

Exhausting Possibilities

So you might ask: Even though we can’t as yet figure out which Calabi-Yau shape string theory selects, does any choice yield physical properties that agree with what we observe? In other words, if we were to work out the corresponding physical properties associated with each and every CalabiYau shape and collect them in a giant catalog, would we find any that match reality? This is an important question, but, for two main reasons, it is also a hard one to answer fully.

A sensible start is to focus only on those Calabi-Yau shapes that yield three families. This cuts down the list of viable choices considerably, although many still remain. In fact, notice that we can deform a multihandled doughnut from one shape to a slew of others—an infinite variety, in fact—without changing the number of holes it contains. In Figure 9.2 we illustrate one such deformation of the bottom shape from Figure 9.1. In much the same way, we can start with a three-holed Calabi-Yau space and smoothly deform its shape without changing the number of holes, again through what amounts to an infinite sequence of shapes. (When we mentioned earlier that there were tens of thousands of Calabi-Yau shapes, we were already grouping together all those shapes that can be changed into one